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1.3.8 Submultiplicative norms

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3:29 / 3:29

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and L_B

does times the maximum stretching that L_A does. How do we capture that? We

would like our norm to have the property that the norm of A times B

is no more than the norm of A times the norm of B . Alright. The norm of B

measures the maximal stretching that B can do. The norm of A measures the maximal

stretching that A can do. Multiply them together, we want something bigger

than the net stretching that the matrix that results from multiplying A times B

does. Okay, now it turns

Video

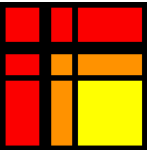
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Reading assignment

1/1 point (graded)



Read Unit 1.3.8 of the notes. [[LINK](#)]

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The solutions in the link provided do not render correctly :(Any suggestions?

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Homework 1.3.8.1

1/1 point (graded)

The Frobenius norm is subordinate to the vector 2-norm.

ALWAYS

 Answer: ALWAYS

For full solution see notes: [[LINK](#)]

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 Answers are displayed within the problem

Homework 1.3.8.2

1/1 point (graded)
Show that $\|Ax\|_\mu \leq \|A\|_{\mu,\nu} \|x\|_\nu$.

 done



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Homework 1.3.8.3

1/1 point (graded)
Show that $\|AB\|_\mu \leq \|A\|_{\mu,\nu} \|B\|_{\nu,\mu}$.

 done



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 Correct (1/1 point)

Homework 1.3.8.4

1/1 point (graded)
Show that the Frobenius norm, $\|\cdot\|_F$, is submultiplicative.

 done



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 Correct (1/1 point)

Homework 1.3.8.5

1/1 point (graded)
For $A \in \mathbb{C}^{m \times n}$ define

$$\|A\| = \max_{i=0}^{m-1} \max_{j=0}^{n-1} |\alpha_{i,j}|.$$

This is a norm.

TRUE

✔ Answer: TRUE

This is a consistent norm.

TRUE

✔ Answer: TRUE

For full solution see notes: [[LINK](#)]

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Answers are displayed within the problem

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